Fair Trade Supply Chain Tracker

Comprehensive Database Output Report

CS251M — Database Systems Project

Generated: April 12, 2026 at 02:55 PM

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PostgreSQL 16 | SQL (CTEs, Triggers, Views, Functions) | Python 3 | Graphviz

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1 Schema Overview — Tables, Triggers, Views, Functions

Overview of all database objects created by setup.sql — 5 tables, 5 triggers, 6 views, and stored functions.

```
SELECT table_name FROM information_schema.tables WHERE table_schema = 'public' AND table_type = 'BASE TABLE' ORDER BY table_name;
```

table_name
audit_log
batch_relations
batches
certifications
stakeholders

(5 rows)

```
SELECT trigger_name, event_manipulation, event_object_table, action_timing FROM information_schema.triggers WHERE trigger_schema = 'public' ORDER BY event_object_table, action_timing;
```

trigger_name	event_manipulation	event_object_table	action_timing
audit_immutability_trigger	DELETE	audit_log	BEFORE
audit_immutability_trigger	UPDATE	audit_log	BEFORE
audit_insert_trigger	INSERT	batches	AFTER
audit_update_trigger	UPDATE	batches	AFTER
certification_trigger	INSERT	batches	BEFORE
transfer_certification_trigger	UPDATE	batches	BEFORE

(6 rows)

```
SELECT table_name AS view_name FROM information_schema.views WHERE table_schema = 'public' ORDER BY table_name;
```

view_name
v_audit_trail
v_batch_details
v_compliance_overview
v_final_products
v_raw_materials
v_supply_chain_summary

(6 rows)

2 Stakeholders & Certifications Data

17 stakeholders across 3 supply chains (Coffee, Cocoa, Cotton) plus 2 edge-case stakeholders for trigger testing.

```
SELECT id, name, role, certification_status, certification_expiry FROM stakeholders ORDER BY id;
```

id	name	role	certification_status	certification_expiry
1	Green Valley Farm	farmer	True	2030-01-01
2	Hilltop Coffee Farm	farmer	True	2030-01-01
3	Coffee Processors Ltd	processor	True	2030-01-01
4	Global Coffee Distributors	distributor	True	2030-01-01
5	City Coffee Retail	retailer	True	2030-01-01
6	Cocoa Farms West	farmer	True	2030-01-01
7	Rainforest Cocoa Ltd	farmer	True	2030-01-01
8	Cocoa Processing Hub	processor	True	2030-01-01
9	Chocolate Distributors Inc	distributor	True	2030-01-01
10	Sweet Retail Store	retailer	True	2030-01-01
11	Cotton Growers Union	farmer	True	2030-01-01
12	Organic Cotton Farm	farmer	True	2030-01-01
13	Textile Processing Ltd	processor	True	2030-01-01
14	Fabric Distributors	distributor	True	2030-01-01
15	Clothing Retail Hub	retailer	True	2030-01-01
16	Expired Farm	farmer	True	2020-01-01
17	Uncertified Retailer	retailer	False	NULL

(17 rows)

```
SELECT s.name, s.role, c.certifying_body, c.issue_date, c.expiry_date, c.is_active
FROM certifications c JOIN stakeholders s ON c.stakeholder_id = s.id ORDER BY s.id;
```

name	role	certifying_body	issue_date	expiry_date	is_active
Green Valley Farm	farmer	FairTrade	2025-01-01	2030-01-01	True
Hilltop Coffee Farm	farmer	Organic	2025-01-01	2030-01-01	True
Coffee Processors Ltd	processor	Food Safety	2025-01-01	2030-01-01	True
Global Coffee Distributors	distributor	Export Cert	2025-01-01	2030-01-01	True

City Coffee Retail	retailer	Retail Cert	2025-01-01	2030-01-01	True
Cocoa Farms West	farmer	FairTrade	2025-01-01	2030-01-01	True
Rainforest Cocoa Ltd	farmer	Organic	2025-01-01	2030-01-01	True
Cocoa Processing Hub	processor	Processing Cert	2025-01-01	2030-01-01	True
Chocolate Distributors Inc	distributor	Export Cert	2025-01-01	2030-01-01	True
Sweet Retail Store	retailer	Retail Cert	2025-01-01	2030-01-01	True
Cotton Growers Union	farmer	Organic Cotton	2025-01-01	2030-01-01	True
Organic Cotton Farm	farmer	Eco Cert	2025-01-01	2030-01-01	True
Textile Processing Ltd	processor	Textile Cert	2025-01-01	2030-01-01	True
Fabric Distributors	distributor	Distribution Cert	2025-01-01	2030-01-01	True
Clothing Retail Hub	retailer	Retail Cert	2025-01-01	2030-01-01	True

(15 rows)

3 All Product Batches with Owner Details

24 product batches across Coffee (100-107), Cocoa (200-207), and Cotton (300-307) chains.

```
SELECT b.id, b.product_name, b.quantity, b.unit, s.name AS owner, s.role
FROM batches b JOIN stakeholders s ON b.current_owner_id = s.id ORDER BY b.id;
```

id	product_name	quantity	unit	owner	role
100	Coffee Raw A	1000	kg	Green Valley Farm	farmer
101	Coffee Raw B	800	kg	Hilltop Coffee Farm	farmer
102	Coffee Proc A1	600	kg	Coffee Processors Ltd	processor
103	Coffee Proc A2	400	kg	Coffee Processors Ltd	processor
104	Coffee Proc B1	800	kg	Coffee Processors Ltd	processor
105	Coffee Export	1800	kg	Global Coffee Distributors	distributor
106	Coffee Retail 1	900	kg	City Coffee Retail	retailer
107	Coffee Retail 2	900	kg	City Coffee Retail	retailer
200	Cocoa Raw A	900	kg	Cocoa Farms West	farmer
201	Cocoa Raw B	700	kg	Rainforest Cocoa Ltd	farmer
202	Cocoa Proc A1	500	kg	Cocoa Processing Hub	processor
203	Cocoa Proc A2	400	kg	Cocoa Processing Hub	processor
204	Cocoa Proc B1	700	kg	Cocoa Processing Hub	processor
205	Cocoa Export	1600	kg	Chocolate Distributors Inc	distributor
206	Chocolate Batch 1	800	kg	Sweet Retail Store	retailer
207	Chocolate Batch 2	800	kg	Sweet Retail Store	retailer
300	Cotton Raw A	1200	kg	Cotton Growers Union	farmer
301	Cotton Raw B	900	kg	Organic Cotton Farm	farmer
302	Cotton Proc A1	700	kg	Textile Processing Ltd	processor
303	Cotton Proc A2	500	kg	Textile Processing Ltd	processor
304	Cotton Proc B1	900	kg	Textile Processing Ltd	processor
305	Fabric Export	2100	kg	Fabric Distributors	distributor
306	Clothing Batch 1	1000	kg	Clothing Retail Hub	retailer
307	Clothing Batch 2	1100	kg	Clothing Retail Hub	retailer

(24 rows)

4 Supply Chain Relationships (Batch Relations)

Parent-child relationships forming a Directed Acyclic Graph (DAG) — models splitting and merging of batches.

```
SELECT br.parent_batch_id, pb.product_name AS parent_product,
       br.child_batch_id, cb.product_name AS child_product
FROM batch_relations br
JOIN batches pb ON br.parent_batch_id = pb.id
JOIN batches cb ON br.child_batch_id = cb.id
ORDER BY br.parent_batch_id, br.child_batch_id;
```

parent_batch_id	parent_product	child_batch_id	child_product
100	Coffee Raw A	102	Coffee Proc A1
100	Coffee Raw A	103	Coffee Proc A2
101	Coffee Raw B	104	Coffee Proc B1
102	Coffee Proc A1	105	Coffee Export
103	Coffee Proc A2	105	Coffee Export
104	Coffee Proc B1	105	Coffee Export
105	Coffee Export	106	Coffee Retail 1
105	Coffee Export	107	Coffee Retail 2
200	Cocoa Raw A	202	Cocoa Proc A1
200	Cocoa Raw A	203	Cocoa Proc A2
201	Cocoa Raw B	204	Cocoa Proc B1
202	Cocoa Proc A1	205	Cocoa Export
203	Cocoa Proc A2	205	Cocoa Export
204	Cocoa Proc B1	205	Cocoa Export
205	Cocoa Export	206	Chocolate Batch 1
205	Cocoa Export	207	Chocolate Batch 2
300	Cotton Raw A	302	Cotton Proc A1
300	Cotton Raw A	303	Cotton Proc A2
301	Cotton Raw B	304	Cotton Proc B1
302	Cotton Proc A1	305	Fabric Export
303	Cotton Proc A2	305	Fabric Export
304	Cotton Proc B1	305	Fabric Export

305	Fabric Export	306	Clothing Batch 1
305	Fabric Export	307	Clothing Batch 2

(24 rows)

5 Backward Lineage Tracing — Coffee Retail to Origin

Traces 'Coffee Retail 1' (batch 106) backward through each supply chain stage to its raw material origins using a Recursive CTE.

```
WITH RECURSIVE lineage AS (
  SELECT b.id, b.product_name, s.name AS owner, s.role, 0 AS depth
  FROM batches b JOIN stakeholders s ON b.current_owner_id = s.id
  WHERE b.id = 106
  UNION ALL
  SELECT parent.id, parent.product_name, ps.name, ps.role, l.depth + 1
  FROM batches parent
  JOIN stakeholders ps ON parent.current_owner_id = ps.id
  JOIN batch_relations br ON br.parent_batch_id = parent.id
  JOIN lineage l ON l.id = br.child_batch_id
)
SELECT * FROM lineage ORDER BY depth DESC;
```

id	product_name	owner	role	depth
101	Coffee Raw B	Hilltop Coffee Farm	farmer	3
100	Coffee Raw A	Green Valley Farm	farmer	3
100	Coffee Raw A	Green Valley Farm	farmer	3
104	Coffee Proc B1	Coffee Processors Ltd	processor	2
102	Coffee Proc A1	Coffee Processors Ltd	processor	2
103	Coffee Proc A2	Coffee Processors Ltd	processor	2
105	Coffee Export	Global Coffee Distributors	distributor	1
106	Coffee Retail 1	City Coffee Retail	retailer	0

(8 rows)

6 Forward Tracing — Impact Analysis from Raw Material

Simulates a recall: traces 'Cocoa Raw A' (batch 200) forward to find ALL downstream products affected.

```
WITH RECURSIVE trace AS (
    SELECT b.id, b.product_name, s.name AS owner, s.role, 0 AS depth
    FROM batches b JOIN stakeholders s ON b.current_owner_id = s.id
    WHERE b.id = 200
    UNION ALL
    SELECT child.id, child.product_name, cs.name, cs.role, t.depth + 1
    FROM batches child
    JOIN stakeholders cs ON child.current_owner_id = cs.id
    JOIN batch_relations br ON br.child_batch_id = child.id
    JOIN trace t ON t.id = br.parent_batch_id
)
SELECT * FROM trace ORDER BY depth;
```

id	product_name	owner	role	depth
200	Cocoa Raw A	Cocoa Farms West	farmer	0
202	Cocoa Proc A1	Cocoa Processing Hub	processor	1
203	Cocoa Proc A2	Cocoa Processing Hub	processor	1
205	Cocoa Export	Chocolate Distributors Inc	distributor	2
205	Cocoa Export	Chocolate Distributors Inc	distributor	2
206	Chocolate Batch 1	Sweet Retail Store	retailer	3
206	Chocolate Batch 1	Sweet Retail Store	retailer	3
207	Chocolate Batch 2	Sweet Retail Store	retailer	3
207	Chocolate Batch 2	Sweet Retail Store	retailer	3

(9 rows)

7 Certification Trigger — Blocking Uncertified Stakeholder

The BEFORE INSERT trigger prevents batch creation if the stakeholder lacks a valid, active certification. Stakeholder 17 has NO certification.

```
INSERT INTO batches (product_name, quantity, unit, current_owner_id) VALUES ('Illegal Batch', 100, 'kg', 17);
```

✗ ERROR: Stakeholder is not certified or certification expired

8 Certification Trigger — Blocking Expired Certification

Stakeholder 16 ('Expired Farm') has certification_expiry = 2020-01-01. The trigger correctly rejects the transaction.

```
INSERT INTO batches (product_name, quantity, unit, current_owner_id) VALUES ('Expired Cert Batch', 100, 'kg', 16);
```

✗ ERROR: Stakeholder is not certified or certification expired

9 Batch Transfer with Automatic Audit Logging

Demonstrates transfer_batch() function — changes batch ownership, validates certification, and automatically logs to audit trail.

BEFORE Transfer:

id	product_name	current_owner
106	Coffee Retail 1	City Coffee Retail

(1 row)

```
SELECT transfer_batch(106, 4);
```

✓ Batch "Coffee Retail 1" (id=106) transferred from City Coffee Retail to Global Coffee Distributors.

AFTER Transfer:

id	product_name	current_owner
106	Coffee Retail 1	Global Coffee Distributors

(1 row)

Audit Trail (transfer logged automatically):

log_id	action	batch_id	product_name	actor_name	actor_role	timestamp
27	transferred	106	Coffee Retail 1	Global Coffee Distributors	distributor	2026-04-12 14:55:32.719444
7	created	106	Coffee Retail 1	City Coffee Retail	retailer	2026-04-12 14:20:47.103503

(2 rows)

10 Audit Log Immutability — Tamper-Proof Verification

The audit_immutability_trigger prevents any UPDATE or DELETE on audit_log — ensuring a tamper-proof, append-only ledger.

```
UPDATE audit_log SET action = 'hacked' WHERE id = 1;
```

✗ ERROR: Audit log records are immutable and cannot be modified or deleted.

```
DELETE FROM audit_log WHERE id = 1;
```

✗ ERROR: Audit log records are immutable and cannot be modified or deleted.

11 Views — Compliance Overview

Real-time compliance status for every stakeholder — COMPLIANT, EXPIRED, REVOKED, or NO CERTIFICATION.

```
SELECT * FROM v_compliance_overview ORDER BY stakeholder_id;
```

stakeholder_id	name	role	certifying_body	issue_date	expiry_date	is_active	compliance_status
1	Green Valley Farm	farmer	FairTrade	2025-01-01	2030-01-01	True	COMPLIANT
2	Hilltop Coffee Farm	farmer	Organic	2025-01-01	2030-01-01	True	COMPLIANT
3	Coffee Processors Ltd	processor	Food Safety	2025-01-01	2030-01-01	True	COMPLIANT
4	Global Coffee Distributors	distributor	Export Cert	2025-01-01	2030-01-01	True	COMPLIANT
5	City Coffee Retail	retailer	Retail Cert	2025-01-01	2030-01-01	True	COMPLIANT
6	Cocoa Farms West	farmer	FairTrade	2025-01-01	2030-01-01	True	COMPLIANT
7	Rainforest Cocoa Ltd	farmer	Organic	2025-01-01	2030-01-01	True	COMPLIANT
8	Cocoa Processing Hub	processor	Processing Cert	2025-01-01	2030-01-01	True	COMPLIANT
9	Chocolate Distributors Inc	distributor	Export Cert	2025-01-01	2030-01-01	True	COMPLIANT
10	Sweet Retail Store	retailer	Retail Cert	2025-01-01	2030-01-01	True	COMPLIANT
11	Cotton Growers Union	farmer	Organic Cotton	2025-01-01	2030-01-01	True	COMPLIANT
12	Organic Cotton Farm	farmer	Eco Cert	2025-01-01	2030-01-01	True	COMPLIANT
13	Textile Processing Ltd	processor	Textile Cert	2025-01-01	2030-01-01	True	COMPLIANT
14	Fabric Distributors	distributor	Distribution Cert	2025-01-01	2030-01-01	True	COMPLIANT
15	Clothing Retail Hub	retailer	Retail Cert	2025-01-01	2030-01-01	True	COMPLIANT
16	Expired Farm	farmer	NULL	NULL	NULL	NULL	NO CERTIFICATION
17	Uncertified Retailer	retailer	NULL	NULL	NULL	NULL	NO CERTIFICATION

(17 rows)

12 Views — Supply Chain Summary

Aggregated statistics per product supply chain.

```
SELECT * FROM v_supply_chain_summary;
```

supply_chain	total_batches	total_quantity	unique_stakeholders
Cocoa	8	6400	5

Coffee	8	7200	5
Cotton	8	8400	5

(3 rows)

13 Views — Final Retail Products & Raw Materials

v_final_products shows leaf batches (retail-ready). v_raw_materials shows root batches (farm-origin).

Final Retail Products (Leaf Nodes):

batch_id	product_name	amount	retailer
106	Coffee Retail 1	900 kg	City Coffee Retail
107	Coffee Retail 2	900 kg	City Coffee Retail
206	Chocolate Batch 1	800 kg	Sweet Retail Store
207	Chocolate Batch 2	800 kg	Sweet Retail Store
306	Clothing Batch 1	1000 kg	Clothing Retail Hub
307	Clothing Batch 2	1100 kg	Clothing Retail Hub

(6 rows)

Raw Material Sources (Root Nodes):

batch_id	product_name	amount	source_farm
100	Coffee Raw A	1000 kg	Green Valley Farm
101	Coffee Raw B	800 kg	Hilltop Coffee Farm
200	Cocoa Raw A	900 kg	Cocoa Farms West
201	Cocoa Raw B	700 kg	Rainforest Cocoa Ltd
300	Cotton Raw A	1200 kg	Cotton Growers Union
301	Cotton Raw B	900 kg	Organic Cotton Farm

(6 rows)

14 Stored Function — get_backward_lineage()

Wraps recursive CTE into a reusable function. Tracing Chocolate Batch 2 (id=207) back to cocoa farm origins.

```
SELECT * FROM get_backward_lineage(207);
```

batch_id	product_name	owner_name	owner_role	depth
201	Cocoa Raw B	Rainforest Cocoa Ltd	farmer	3
200	Cocoa Raw A	Cocoa Farms West	farmer	3
200	Cocoa Raw A	Cocoa Farms West	farmer	3
204	Cocoa Proc B1	Cocoa Processing Hub	processor	2
202	Cocoa Proc A1	Cocoa Processing Hub	processor	2
203	Cocoa Proc A2	Cocoa Processing Hub	processor	2
205	Cocoa Export	Chocolate Distributors Inc	distributor	1
207	Chocolate Batch 2	Sweet Retail Store	retailer	0

(8 rows)

15 Stored Function — get_forward_trace()

Traces Cotton Raw A (id=300) forward to find all downstream clothing products.

```
SELECT * FROM get_forward_trace(300);
```

batch_id	product_name	owner_name	owner_role	depth
300	Cotton Raw A	Cotton Growers Union	farmer	0
302	Cotton Proc A1	Textile Processing Ltd	processor	1
303	Cotton Proc A2	Textile Processing Ltd	processor	1
305	Fabric Export	Fabric Distributors	distributor	2
305	Fabric Export	Fabric Distributors	distributor	2
306	Clothing Batch 1	Clothing Retail Hub	retailer	3
306	Clothing Batch 1	Clothing Retail Hub	retailer	3
307	Clothing Batch 2	Clothing Retail Hub	retailer	3
307	Clothing Batch 2	Clothing Retail Hub	retailer	3

(9 rows)

16 Stored Function — get_compliance_report()

Full compliance audit — stakeholders 16 and 17 are flagged as non-compliant.

```
SELECT * FROM get_compliance_report();
```

stakeholder_id	stakeholder_name	role	certifying_body	expiry_date	compliance_status
1	Green Valley Farm	farmer	FairTrade	2030-01-01	COMPLIANT
2	Hilltop Coffee Farm	farmer	Organic	2030-01-01	COMPLIANT
3	Coffee Processors Ltd	processor	Food Safety	2030-01-01	COMPLIANT
4	Global Coffee Distributors	distributor	Export Cert	2030-01-01	COMPLIANT
5	City Coffee Retail	retailer	Retail Cert	2030-01-01	COMPLIANT
6	Cocoa Farms West	farmer	FairTrade	2030-01-01	COMPLIANT
7	Rainforest Cocoa Ltd	farmer	Organic	2030-01-01	COMPLIANT
8	Cocoa Processing Hub	processor	Processing Cert	2030-01-01	COMPLIANT
9	Chocolate Distributors Inc	distributor	Export Cert	2030-01-01	COMPLIANT
10	Sweet Retail Store	retailer	Retail Cert	2030-01-01	COMPLIANT
11	Cotton Growers Union	farmer	Organic Cotton	2030-01-01	COMPLIANT
12	Organic Cotton Farm	farmer	Eco Cert	2030-01-01	COMPLIANT
13	Textile Processing Ltd	processor	Textile Cert	2030-01-01	COMPLIANT
14	Fabric Distributors	distributor	Distribution Cert	2030-01-01	COMPLIANT
15	Clothing Retail Hub	retailer	Retail Cert	2030-01-01	COMPLIANT
16	Expired Farm	farmer	NONE	NULL	NO CERTIFICATION
17	Uncertified Retailer	retailer	NONE	NULL	NO CERTIFICATION

(17 rows)

17 Analytical Queries — Aggregates & Statistics

Aggregate and analytical queries — GROUP BY, HAVING, recursive depth, and audit statistics.

Total Quantity per Supply Chain Stage:

```
SELECT s.role, SUM(b.quantity) AS total_quantity, COUNT(b.id) AS batch_count
FROM batches b JOIN stakeholders s ON b.current_owner_id = s.id
GROUP BY s.role ORDER BY CASE s.role WHEN 'farmer' THEN 1 WHEN 'processor' THEN 2
WHEN 'distributor' THEN 3 WHEN 'retailer' THEN 4 END;
```

role	total_quantity	batch_count
farmer	5500	6
processor	5500	9
distributor	5500	3
retailer	5500	6

(4 rows)

Branching Factor — Batches with Most Children:

```
SELECT b.id, b.product_name, COUNT(br.child_batch_id) AS children_count
FROM batches b LEFT JOIN batch_relations br ON b.id = br.parent_batch_id
GROUP BY b.id, b.product_name HAVING COUNT(br.child_batch_id) > 0
ORDER BY children_count DESC;
```

id	product_name	children_count
305	Fabric Export	2
205	Cocoa Export	2
300	Cotton Raw A	2
105	Coffee Export	2
200	Cocoa Raw A	2
100	Coffee Raw A	2
102	Coffee Proc A1	1
103	Coffee Proc A2	1
204	Cocoa Proc B1	1
104	Coffee Proc B1	1
303	Cotton Proc A2	1
202	Cocoa Proc A1	1

301	Cotton Raw B	1
201	Cocoa Raw B	1
304	Cotton Proc B1	1
101	Coffee Raw B	1
302	Cotton Proc A1	1
203	Cocoa Proc A2	1

(18 rows)

Supply Chain Depth per Root Batch:

```
WITH RECURSIVE chain AS (
    SELECT b.id AS root_id, b.product_name AS root_product, b.id, 0 AS depth
    FROM batches b WHERE b.id NOT IN (SELECT child_batch_id FROM batch_relations)
    UNION ALL
    SELECT c.root_id, c.root_product, br.child_batch_id, c.depth + 1
    FROM chain c JOIN batch_relations br ON br.parent_batch_id = c.id
)
SELECT root_id, root_product, MAX(depth) AS max_chain_depth
FROM chain GROUP BY root_id, root_product ORDER BY root_id;
```

root_id	root_product	max_chain_depth
100	Coffee Raw A	3
101	Coffee Raw B	3
200	Cocoa Raw A	3
201	Cocoa Raw B	3
300	Cotton Raw A	3
301	Cotton Raw B	3

(6 rows)

Audit Actions Summary:

```
SELECT action, COUNT(*) AS total FROM audit_log GROUP BY action ORDER BY total DESC;
```

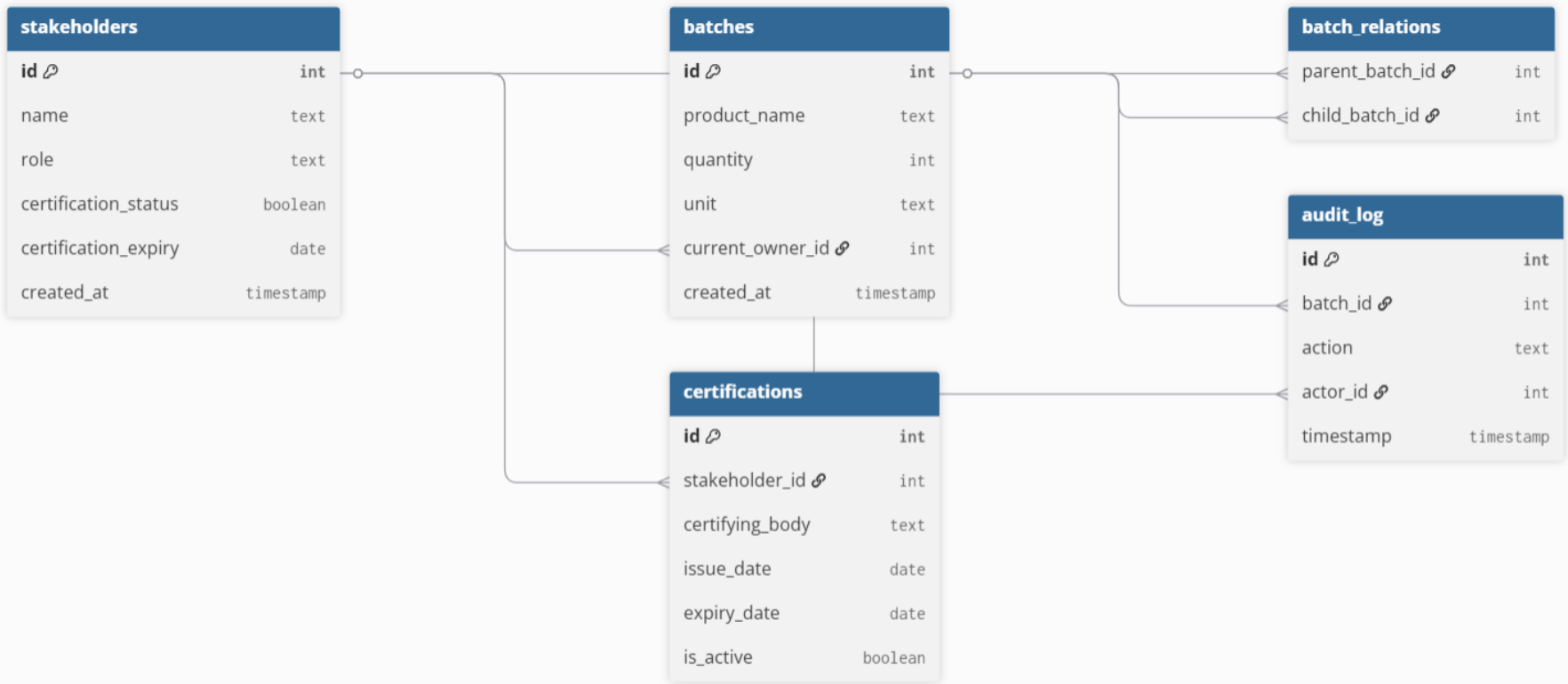
action	total
created	24

(1 row)

18 ER Diagram & Supply Chain Visualization

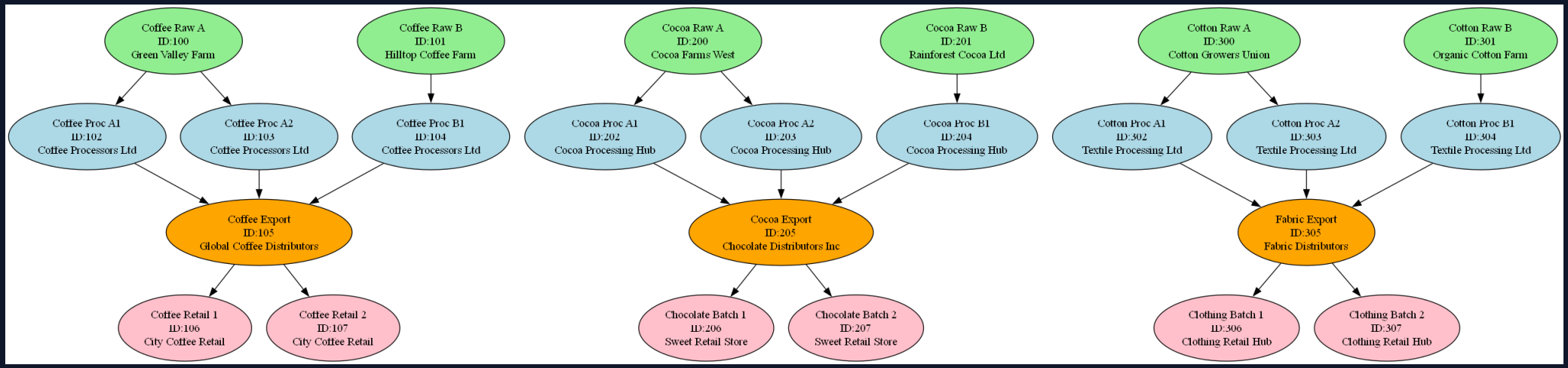
Visual representations of the database schema and supply chain data. Auto-generated using Python + Graphviz.

Entity-Relationship Diagram:



ER Diagram — 5 tables with foreign key relationships

Supply Chain Graph (Generated from Live Data):



Supply Chain DAG — 3 chains with color-coded stakeholder roles